26900063 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Elective (DSE) - MG5DSEBTG303 - MARINE BIOTECHNOLOGY

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

One Word Answer Questions

Answer any **10** questions

Each question carries **2** marks

1. How does marine biotechnology help to control pollution? [K] / [CO1]
2. Identify the enzyme produced when cellulose is used as the substrate. [K] / [CO2]
3. Mention any two methods of preserving marine samples. [U] / [CO3]
4. What is chitin? [K] / [CO2]
5. What is the study of evolutionary relationships using genetic data called? [K] / [CO1]
6. Which organisms are the major primary producers in the ocean? [K] / [CO1]
7. Name one biological interaction among marine microorganisms. [K] / [CO1]
8. Recall what AAS stands for. [K] / [CO3]
9. Site one example for In-situ and Ex-situ conservation. [K] / [CO4]

10. Mention one extreme habitat inhabited by marine bacteria. [K] / [CO1]

11. What process converts sunlight into chemical energy? [K] / [CO1]

12. Define oceanographic sampling. [U] / [CO3]

[2x10 = 20]

Part B

Short Essay Questions

Answer any **4** questions

Each question carries **5** marks

13. Explain the principle and applications of biosensors in marine environmental monitoring. [U] / [CO3]

14. Explain biosorption of heavy metals using marine biomass. [U] / [CO4]

15. Discuss the role of genetic diversity in the adaptation, evolution and survival of marine organisms. [U] / [CO1]

16. Analyse the importance of combining genomics and natural-product chemistry in marine drug discovery. [An] / [CO2]

17. Predict the effect of using closed photobioreactors instead of open ponds on contamination and biomass productivity. [An] / [CO2]

18. Describe the applications of PCR in marine biotechnology. [U] / [CO3]

[5x4 = 20]

Part C

Essay Questions

Answer any **2** questions

Each question carries **15** marks

19. Explain the future prospects of marine biotechnology and its contribution to the blue economy. [U] / [CO4]

20. Describe quality assurance, quality control and contamination prevention during marine sample collection. [U] / [CO3]

21. Identify the important characteristics, diversity, ecological roles, and biotechnological importance of marine microorganisms. [K] / [CO1]

22. Explain how a systematic study of a marine organism can be documented in the form of a scientific study report. [U] / [CO3]

[15x2 = 30]